THE THREE TRIBES OF EMBODIED INTELLIGENCE

Bridging the Disciplinary Divides Between Computer Science, Embedded Systems, and Mechanical Robotics

DISCIPLINE 1: COMPUTER SCIENCE / AI

Machine Learning Perspective

- Loss functions · Transformers · Latent world models
- Python abstractions · Generative policies · Tokenization

DISCIPLINE 2: EMBEDDED SYSTEMS / ECE

Silicon & Firmware Perspective

- Microsecond jitter · DMA memory buses · Interrupt service routines
- Static SRAM (malloc=0) · Watchdogs · AXI QoS priorities

PHYSICAL AI

DISCIPLINE 3: ROBOTICS & MECHANICS

Kinematics & Dynamics Perspective

- Kinetic momentum · Friction cones · Contact compliance
- Stopping distance d_stop · Gearbox backlash · Dynamic stability